

HC-POD CROSS-TRACK TESTING & REFINEMENT EVIDENCE

HealthConnect Experience Lab — Data Analytics Track

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Cross-Track Collaboration: Data Analytics × Data Science

1. Track Collaborated With

Data Science Track — Donald Nwachukwu

2. Project Dependency

The Data Science track used HealthConnect appointment data and analytical findings from the Data Analytics work to investigate factors associated with appointment No-Shows and develop predictive models.

A specific dependency involved the **Lead Time × Previous No-Show** segmentation, which was being used to understand appointment groups with higher observed No-Show Rates.

Because the two tracks reported different rates for a similar high-risk segment, the calculation needed to be tested and validated before the results could be confidently compared or integrated.

3. Component / Output Being Tested

The component tested was the **Lead Time × Previous No-Show segment calculation**.

The Data Science segment was defined as:

- `booking_lead_days > 30`
- `previous_no_shows ≥ 1`

Donald originally reported a **70.5% No-Show Rate** for this segment.

The Data Analytics work had previously reported a **64.1% No-Show Rate** for its own 3+ weeks + previous No-Show segment.

This difference required cross-track validation because the two results appeared to describe a similar appointment group.

4. Information / Output Received

The Data Analytics track received the exact Data Science calculation criteria from Donald:

- **Booking lead time:** $\text{booking_lead_days} > 30$
- **Previous No-Show history:** $\text{previous_no_shows} \geq 1$
- **Original denominator:** Cancelled appointments excluded.

Donald also provided the calculation showing:

$$689 \text{ No-Shows} \div 977 \text{ eligible appointments} = 70.52\%$$

He subsequently recalculated the same segment with Cancelled appointments included:

$$689 \text{ No-Shows} \div 1,015 \text{ total appointments} = 67.88\%$$

5. Information / Output Provided

The Data Analytics track requested clarification of the denominator and independently reproduced the Data Science calculation using the HealthConnect Data Analytics dataset.

The independent test used the exact Data Science segment definition:

$\text{booking_lead_days} > 30$

AND

$\text{previous_no_shows} \geq 1$

Both denominator definitions were tested:

1. Cancelled appointments excluded.
 2. Cancelled appointments included.
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6. Testing Activity Completed

The following cross-track testing process was completed:

Step 1 — Identify the discrepancy

The Data Analytics and Data Science results showed different reported No-Show Rates for a similar Lead Time × Previous No-Show segment.

Step 2 — Obtain the exact Data Science definition

The Data Science track provided the exact lead-time threshold, previous No-Show condition, and denominator used in its calculation.

Step 3 — Independently reproduce the calculation

The Data Analytics track applied the exact Data Science criteria to its own dataset.

Step 4 — Test both denominator definitions

The calculation was performed with Cancelled appointments excluded and then with Cancelled appointments included.

Step 5 — Compare the results

The independently calculated results matched the Data Science figures exactly.

7. Issue / Finding Identified

The testing identified that the two analyses did not use identical definitions.

There were **two methodological differences**:

Lead-time threshold

- Data Analytics: $\text{booking_lead_days} \geq 22$ for the 3+ weeks segment.
- Data Science: $\text{booking_lead_days} > 30$.

Denominator definition

- Data Analytics: Cancelled appointments included.
- Data Science original calculation: Cancelled appointments excluded.

Therefore, the two reported percentages were not directly comparable without aligning the segment criteria and denominator.

8. Refinement / Action Taken

The analytical definitions were explicitly documented so that future cross-track comparisons would state:

- The booking lead-time threshold.
- The previous No-Show condition.
- Whether Cancelled appointments are included or excluded from the denominator.

The Data Analytics team also independently reproduced the Data Science calculation using the exact Data Science criteria.

9. Retest Result

The cross-track calculation was successfully reproduced.

Calculation Definition

No-Show Rate

>30 days + previous No-Show, Cancelled excluded **70.52%**

>30 days + previous No-Show, Cancelled included **67.88%**

The segment contained **1,015 appointments** when Cancelled appointments were included.

The independent Data Analytics calculation matched the Data Science calculation exactly under the same definition.

Retest Result: PASS

10. What Changed as a Result

The main change was improved **definition alignment and documentation** between the two tracks.

Before testing, the difference between **64.1% and 70.5%** could appear to be a disagreement between the two analyses.

After testing, it was established that the analyses were using different:

- Lead-time thresholds.
- Denominator definitions.

The Data Science calculation was independently reproduced, and the criteria for comparing segment rates were made explicit.

11. Evidence

The following evidence supports the cross-track testing activity:

- Data Science communication containing the exact segment definition.
- Data Science calculation/output showing **70.52%**.

- Data Science retest showing **67.88%** with Cancelled appointments included.
 - Data Analytics Python code independently reproducing both calculations.
 - Python output showing:
 - **70.52%** excluding Cancelled.
 - **67.88%** including Cancelled.
 - **1,015** total appointments in the segment.
 - Week 7 testing record documenting the cross-track validation.
 - Week 7 notebook section documenting the segment-definition clarification.
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12. How This Improved the Overall HealthConnect Solution

This cross-track testing improved the HealthConnect solution by ensuring that analytical results shared between Data Analytics and Data Science are based on clearly defined and reproducible criteria.

The activity also demonstrated that a reported metric should not be compared across tracks without checking:

- Segment definitions.
- Thresholds.
- Outcome categories.
- Denominator rules.

The validated calculation provides a reliable reference for future Data Science modelling, Data Analytics reporting, and cross-track discussions.

It also reduced the risk of different HealthConnect components using apparently similar but technically different definitions.

Cross-Track Testing Summary

The activity followed the required **Test → Finding → Action → Retest → Validated Improvement** process:

Stage	Result
Test	Independently reproduced the Data Science Lead Time $\times$ Previous No-Show segment
Finding	Different lead-time thresholds and denominator definitions were being used
Action	Exact criteria and denominator rules were documented
Retest	Data Science results were independently reproduced
Validated Improvement	Cross-track segment definitions are now clearer and comparable only when the criteria are aligned

Overall Cross-Track Result

PASS — Cross-track calculation independently validated and analytical definition differences documented.